

Three Experiments for Rigorous University Replication

A concept note on testing extraordinary claims about consciousness under clean, ethical, preregistered, and publishable conditions.

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Purpose	Rough proposals and indicative budgets to support early conversations with university partners.
Status	Draft for discussion — figures are order-of-magnitude estimates.

Framing. All three studies are designed as *replication / falsification* studies. The aim is not to prove extraordinary claims, but to test them under conditions that are publishable even when the result is negative. A clean null result is a valuable scientific contribution; a positive result would be highly significant.

Executive Summary

This note outlines three candidate experiments that could anchor a university research agenda on altered states of consciousness, neurophenomenology, and the scientific study of extraordinary human experience. Each is summarised below with a core idea, a proposed design, required infrastructure, ethical considerations, and an indicative budget. A comparative budget table and a longer pipeline of ten further candidates are included at the end.

Study	Core question	Indicative budget	Risk / complexity
1. Extended-State DMT (DMTx)	Can the DMT state be safely extended and characterised in real time?	\$450k – \$650k	High — controlled substance, medical
2. Brain-to-Brain Correlation	Do separated, paired subjects show correlated EEG responses?	\$250k – \$350k	Medium — design rigor is everything
3. Trance / Healer Neurophysiology	Do practitioners enter a reproducible, distinct brain state?	\$200k – \$320k	Low–Medium — recruitment-limited

Recommended sequencing. Studies 2 and 3 are lower-cost, lower-risk, and faster to ethics approval; either could serve as a credible first project that demonstrates rigor to partners and funders. Study 1 (DMTx) is the scientific flagship but carries the heaviest regulatory, medical, and budgetary load and is best pursued once institutional relationships and a clinical partner are in place.

Extended-State DMT / DMTx

Core idea

Rather than studying DMT as a very short (5–15 minute) experience, this protocol uses a controlled intravenous loading dose followed by a target-controlled maintenance infusion to **extend and stabilise the DMT state** for an experimentally useful window. This makes it possible to study the phenomenology, neural correlates, and any recurring structures of the experience while collecting real-time physiological and neuroimaging data. The informal DMT / laser-perception idea we discussed would sit inside this framework as a small, blinded perceptual sub-study rather than as the flagship aim.

Proposed design

- **Design:** open-label pharmacokinetic / pharmacodynamic pilot, $n \approx 8$ –12 healthy, psychedelic-experienced volunteers, within-subject dose escalation.
- **Primary outcomes:** safety and tolerability of extended infusion; feasibility of maintaining a stable subjective state at a target plasma level.
- **Secondary outcomes:** EEG signatures (spectral power, complexity / entropy, connectivity); validated phenomenology scales; structured post-session interviews.
- **Optional blinded sub-study:** randomised, blinded perceptual tasks delivered during the stable state to test specific claims under controlled conditions.
- **Analysis:** preregistered, with predefined safety stopping rules and primary endpoints.

Key infrastructure & equipment

- Clinical research unit with anaesthesia-grade monitoring
- Target-controlled IV infusion pump(s)
- Research-grade EEG (optionally MRI-compatible) and physiological monitoring
- GMP-grade DMT and hospital pharmacy handling
- Controlled-substance storage and chain-of-custody

Ethical & safety considerations

- Schedule I / controlled-substance licensing and import permits
- Full IRB / research-ethics board review
- On-site physician and resuscitation capability
- Rigorous screening; cardiac and psychiatric exclusions
- Trained psychological support and integration

Indicative timeline

Approximately 18–24 months: 6–9 months for regulatory, ethics, and drug supply; 3 months setup and staff training; 6–9 months data collection; 3–6 months analysis and write-up.

Indicative budget (rough / general)

Cost category	Estimate (USD)
Regulatory, licensing, controlled-substance & ethics approvals	\$40,000
GMP DMT supply, pharmacy & secure handling	\$60,000

Clinical & medical staff (physician, anaesthesia, nurse, psychiatrist)	\$120,000
Neuroimaging / EEG equipment or scanner access & physiological monitoring	\$80,000
Core research personnel (PI, postdoc, RAs, analyst — ~1 yr)	\$180,000
Participant screening, compensation & facility costs	\$40,000
Data analysis, computing, dissemination & contingency	\$30,000
Indicative total	≈ \$450,000 – \$650,000

Key references & links

- Gallimore & Strassman (2016), *Frontiers in Pharmacology*
- Extended DMT study (PMC10851633)
- Continuous DMT infusion study (PMC12032411)

Brain-to-Brain Correlation / “Transferred Potential” Replication

Core idea

Inspired by Jacobo Grinberg’s controversial work on “transferred potentials,” in which two people are placed in separate rooms, one receives a stimulus, and the other’s brain activity is monitored for correlated responses. The claim is extraordinary — which is exactly why it deserves a **modern, rigorously controlled replication**. The scientific value here lies almost entirely in design quality: double-blinding, shielding, randomised timing, and fully automated, preregistered analysis remove the loopholes that made the original work impossible to interpret.

Proposed design

- **Design:** paired-subject EEG study, $\approx$ 25–30 pairs (emotionally bonded and stranger pairs), “sender” and “receiver” in separate electromagnetically shielded rooms.
- **Stimulus:** randomised, computer-triggered sensory stimuli to the sender at times unknown to the receiver and to all experimenters present.
- **Primary outcome:** preregistered test for time-locked correlated activity in the receiver’s EEG during sender-stimulation vs. matched null windows.
- **Controls:** sham/no-sender blocks, sensor-level artifact rejection, and analyst blinding to condition labels.
- **Statistics:** preregistered effect-size threshold, Bayesian + frequentist criteria; a well-powered null is a publishable outcome.

Key infrastructure & equipment

- Two research-grade EEG (or EEG/MEG) systems, time-synchronised
- Electromagnetically shielded / Faraday-isolated rooms
- Hardware trigger & precision timing infrastructure
- Automated, preregistered analysis pipeline
- Audio/RF isolation verification equipment

Ethical & safety considerations

- IRB / ethics approval (low physical risk)
- Informed consent; right to withdraw
- Pre-registration on OSF / AsPredicted before data collection
- Independent verification of room isolation
- Open data and analysis code for full transparency

Indicative timeline

Approximately 12–15 months: 2–3 months ethics and preregistration; 2 months setup and isolation testing; 5–6 months paired data collection; 3–4 months analysis and write-up.

Indicative budget (rough / general)

Cost category	Estimate (USD)
Research-grade EEG/MEG equipment (purchase or rental)	\$80,000
Shielded-room setup or facility access & isolation testing	\$30,000

Core personnel (PI part-time, postdoc, 2 RAs, analyst — ~1 yr)	\$160,000
Participant recruitment & compensation (~60 participants)	\$20,000
Preregistration, replication design & statistical consulting	\$15,000
Computing, open-data hosting & dissemination	\$15,000
Indicative total	≈ \$250,000 – \$350,000

Key references & links

- [Jacobo Grinberg — background & publication reference \(Wikipedia, ES\)](#)
- [Related separated-subject EEG study \(PubMed 12972348\)](#)

Shamans / Healers / Trance Practitioners under EEG or fMRI

Core idea

Study shamans, healers, meditators, or trance practitioners while they enter their claimed healing or altered state, and compare their brain activity and physiology against matched controls and against their own ordinary rest, imagination, and meditation. Crucially, this study does **not** need to begin by claiming that “healing energy” is real. The first, stronger scientific question is simply: *do trained practitioners enter a reproducible neurophysiological state that is measurably different from ordinary baselines?* If yes, a later phase can test whether that state has any measurable effect on recipients.

Proposed design

- **Design:** within-subject, multi-condition comparison (trance vs. rest vs. imagery vs. meditation), ≈ 15–25 experienced practitioners + matched controls.
- **Primary outcome:** reproducible, practitioner-specific EEG/fMRI signature that discriminates the trance state from control conditions above chance.
- **Secondary outcomes:** autonomic measures (HRV, electrodermal activity, respiration); state classification with cross-validated machine learning.
- **Phase 2 (optional):** blinded recipient study testing for any physiological or perceptual effect on a second person.
- **Analysis:** preregistered classification accuracy and effect-size thresholds.

Key infrastructure & equipment

- Research-grade EEG (and/or fMRI scanner access)
- Autonomic / peripheral physiology recording (HRV, EDA)
- Synchronised audio-video for state annotation
- Analysis workstation & ML pipeline
- Quiet, controlled recording environment

Ethical & safety considerations

- IRB / ethics approval; culturally respectful protocol
- Community consultation & fair compensation for practitioners
- Informed consent honouring practitioners' traditions
- Clear non-medical-claim framing in all materials
- Pre-registration of hypotheses and analysis

Indicative timeline

Approximately 12–18 months: 2–3 months ethics, community engagement, and recruitment; 5–7 months data collection (recruitment-paced); 3–5 months analysis and write-up.

Indicative budget (rough / general)

Cost category	Estimate (USD)
EEG equipment and/or fMRI scanner time (~40 sessions)	\$60,000
Core personnel (postdoc, RAs, data analyst — ~1 yr)	\$150,000
Practitioner recruitment, travel & compensation	\$35,000

Peripheral physiology hardware (HRV, EDA, respiration)	\$15,000
Ethics, preregistration, dissemination & contingency	\$20,000
Indicative total	≈ \$200,000 – \$320,000

Key references & links

- Cognitive / shamanic-style trance research ([arXiv 2509.19254](#))
- Trance & neurophysiology — general background ([Wikipedia](#))

Comparative Budget Overview

The figures below are **rough, order-of-magnitude estimates** for self-contained pilot studies. Real costs depend heavily on the host institution: where a university already owns EEG/MRI infrastructure, shielded rooms, or a clinical research unit, equipment and facility lines can fall substantially, since these become shared rather than dedicated costs.

Study	Low	High	Typical duration	First-mover?
1 • Extended-State DMT (DMTx)	\$450k	\$650k	18–24 months	Flagship — later phase
2 • Brain-to-Brain Correlation	\$250k	\$350k	12–15 months	Strong candidate
3 • Trance / Healer Neurophysiology	\$200k	\$320k	12–18 months	Strong candidate
Combined programme (all three)	\$900k	\$1.32M	~2–3 years phased	—

Budgets assume external dedicated provision of major equipment. With shared university infrastructure, expect roughly 20–40% lower totals on Studies 2 and 3.

Cross-Cutting Principles

Replication, not advocacy. Every study is framed to be publishable whether the result is positive or null. We are testing claims, not defending them.

Preregistration by default. Hypotheses, sample sizes, and analysis plans are registered before data collection on OSF or a comparable platform, protecting against bias and post-hoc storytelling.

Ethics first. Each study is designed to be ethically approvable at a mainstream university, with appropriate review, consent, safety, and (for Study 3) cultural respect.

Open and transparent. Open data and analysis code wherever ethically possible, so results can be independently verified — the single most important safeguard for controversial questions.

Broader Pipeline — Ten Further Candidates

Beyond the three priority studies, the following are strong additional replication candidates in consciousness and altered-states research, available to build out a longer programme:

1. Real-time communication with lucid dreamers during REM sleep
2. Psilocybin and increased brain-state repertoire
3. Ayahuasca and brain entropy / network reorganization
4. Expert meditators and high-amplitude gamma synchrony
5. Hypnosis and changes in pain, agency, and self-perception
6. Breathwork-induced non-ordinary states without psychedelics
7. Ganzfeld / sensory deprivation and internally generated perception
8. Near-death experience research and end-of-life brain activity
9. Anesthesia and perturbational complexity as a measure of consciousness
10. Virtual-reality body-transfer or out-of-body illusion experiments

Prepared by Ney Torres · 22 June 2026 · Draft concept note for discussion with university partners. All budget figures are indicative and subject to refinement with a host institution.